

Zachary Kralec

zkralec@icloud.com • [Development & Work Portfolio](#) • [GitHub](#)

Software Engineer | AI & Automation Systems | Full-Stack + Infrastructure

SELECTED PROJECT

Mission Control – AI Automation Platform | Python, FastAPI, Redis, OpenAI API, React, TypeScript

- Architected and built a full-stack AI automation platform to orchestrate multi-step workflows with state tracking, retries, and human in the loop review
- Designed distributed task pipelines with idempotent execution, failure recovery, and real-time monitoring
- Integrated LLM decision systems (classification, ranking, evaluation) directly into production workflows
- Built a React dashboard to monitor task execution, system health, and AI usage across all pipelines
- Automated data pipelines for scraping, ranking, and alerting on high value signals

EXPERIENCE

Resource Management Concepts Inc. | Lexington Park, MD

Software & Automation Engineer (Corporate IT): September 2025 – Present

- Designed and deployed automation systems for endpoint auditing across 100+ devices, generating structured and informational security reports and anomaly detection signals
- Built resumable automation workflows to replace manual audit processes, improving reliability and overhead
- Developed internal tools for identity management, access provisioning, and system diagnostics
- Engineered rule-based email filtering systems (regex + logic) to improve phishing email identification and capture rate
- Supported Linux based virtualization environments (Proxmox), including deployment, configuration, and troubleshooting

Resource Management Concepts Inc. | Lexington Park, MD

Automation Engineering Intern (Corporate IT): June 2025 - August 2025

- Automated routine system administration tasks using PowerShell, reducing manual workload for account management and system maintenance
- Diagnosed and resolved infrastructure issues across Windows servers and endpoints
- Contributed to management of a clustered virtualization environment (Proxmox), improving deployment workflows

LTN Global Communications Inc. | Columbia, MD

Software Engineering Intern (QA Automation): May 2024 - August 2024

- Built Python automation tools to help streamline video testing workflows, while reducing execution time by ~50%
- Developed standardized test output systems to improve debugging and engineering visibility
- Maintained and debugged CI/CD pipelines within Gitlab, helping to resolve deployment and integration issues
- Identified and resolved performance issues in video encoding/decoding and processing systems

Sinclair Inc. | Hunt Valley, MD

Business Systems Analyst Intern — June 2023 - August 2023

- Led stakeholder meetings to gather requirements and translate business needs into technical solutions
- Delivered a functional prototype for a centralized product training platform, improving internal knowledge access
- Collaborated directly with business stakeholders to identify pain points and translate needs into 20+ ServiceNow enhancements, reducing resolution time.

SKILLS

- **Languages:** Python, JavaScript, SQL, C/C++, Java, Bash
- **Frameworks:** React, Node.js, FastAPI, Flask
- **AI/Data:** LLM workflows, OpenAI API, Langchain, scikit-learn, Pandas
- **Systems:** Redis, Docker, Linux, REST APIs, Proxmox
- **Cloud:** AWS, Azure, GCP, Firebase
- **Tools:** Git, Kubernetes, PowerShell, PDQ, ServiceNow

EDUCATION

St. Mary's College of Maryland | Saint Mary's City, MD

Bachelor of Science in Computer Science, Minor in Business | GPA: 3.54

Awards & Honors: Dean's List (5x), Senior Scholar Athlete, Chi Alpha Sigma